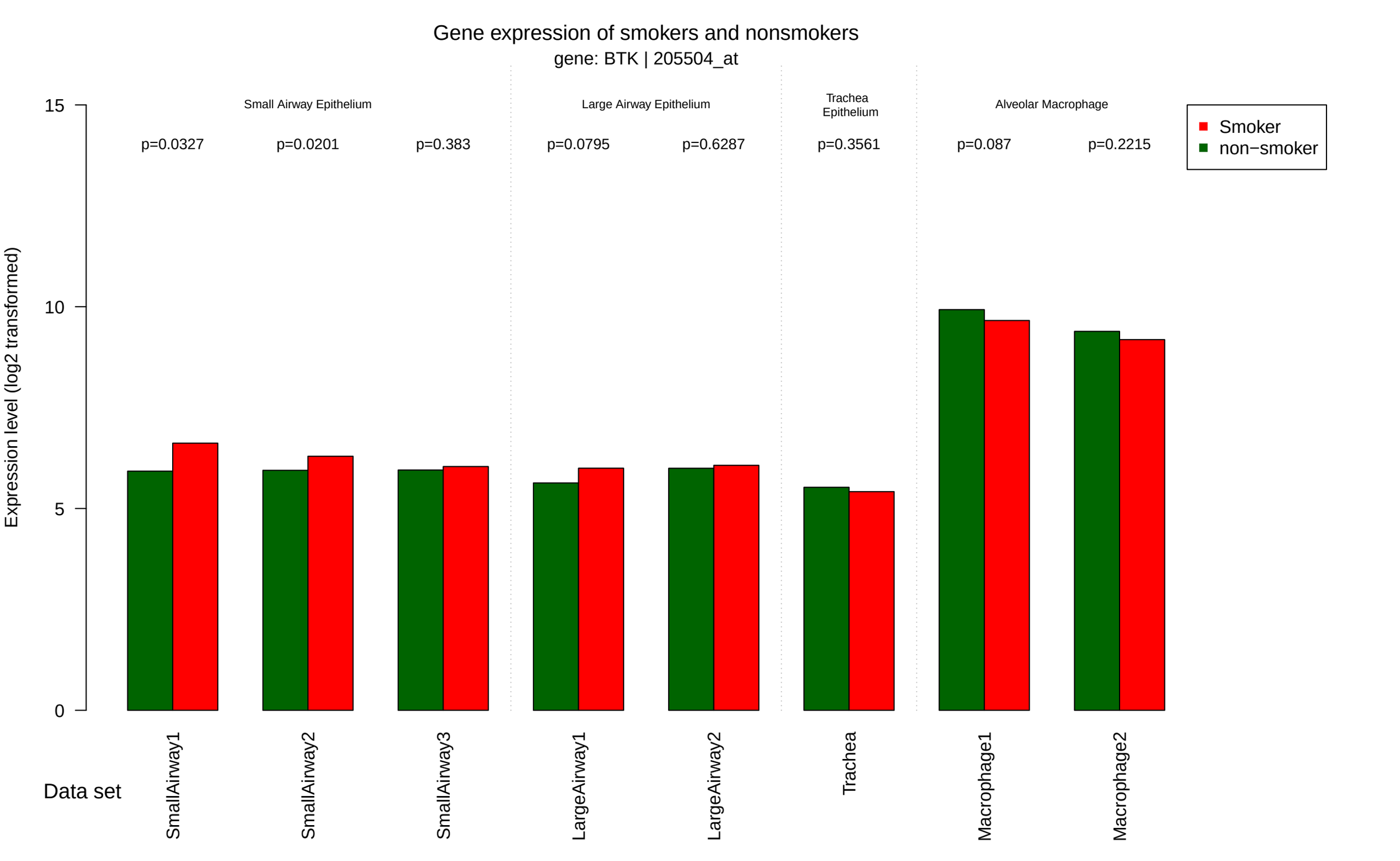
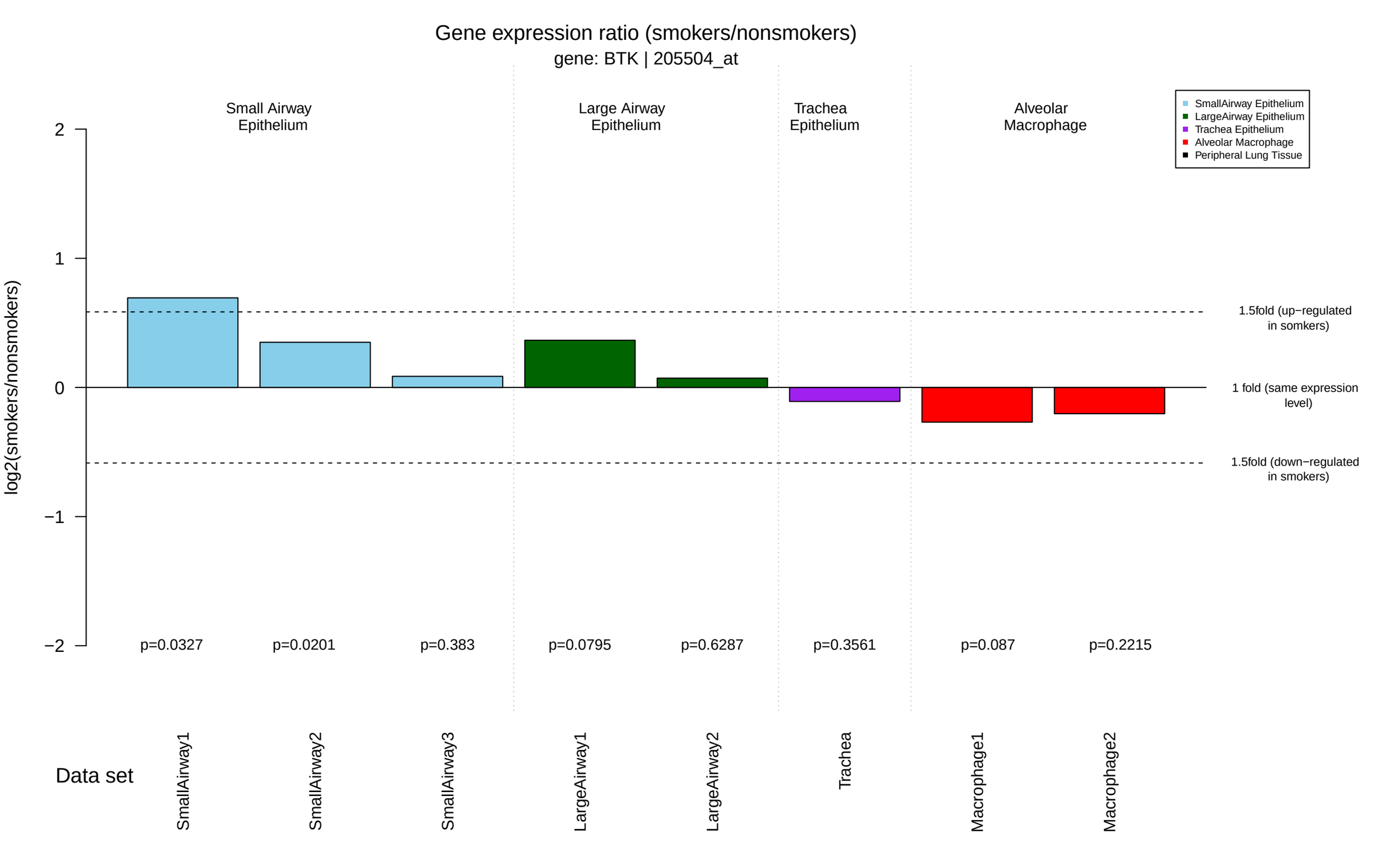
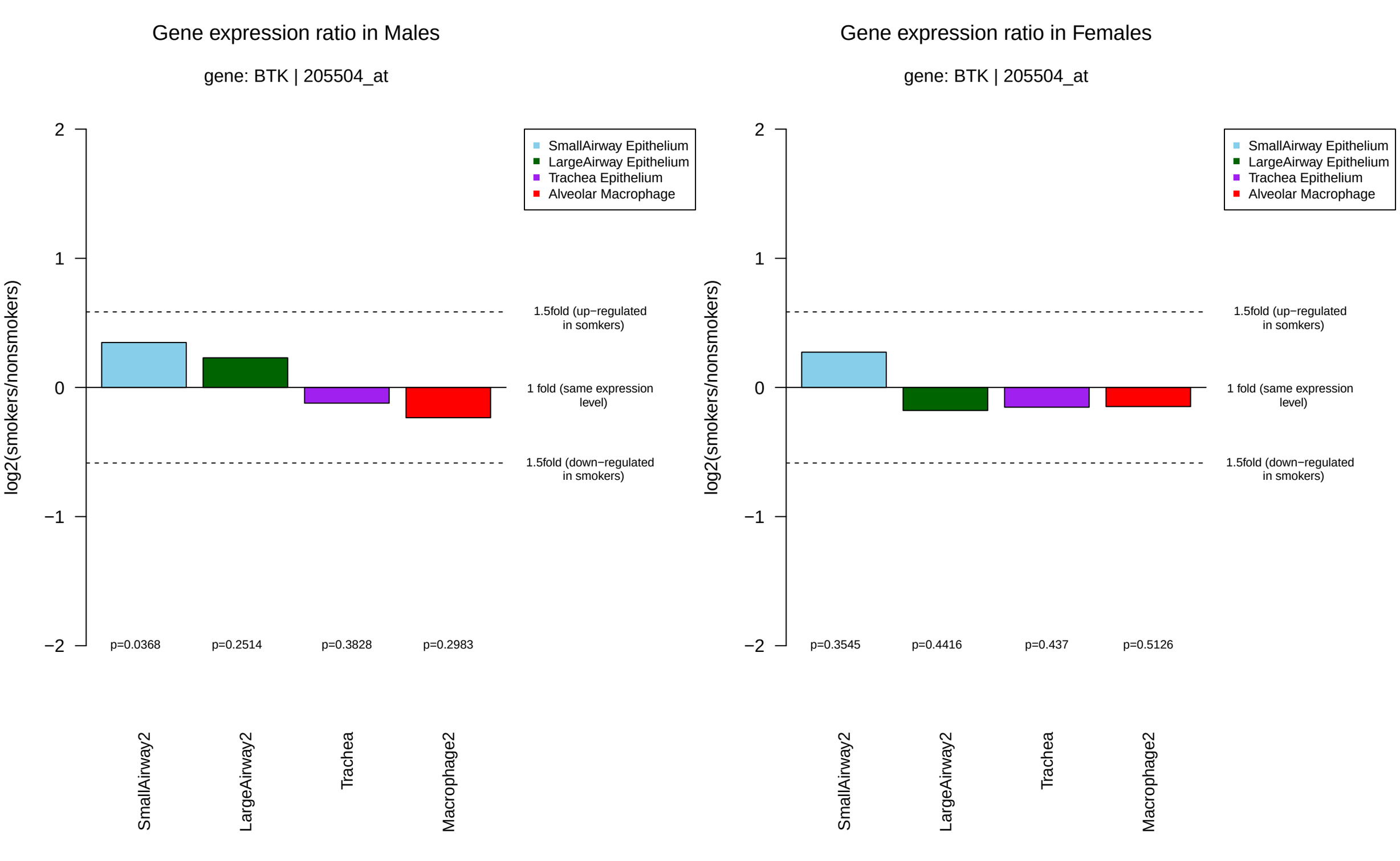
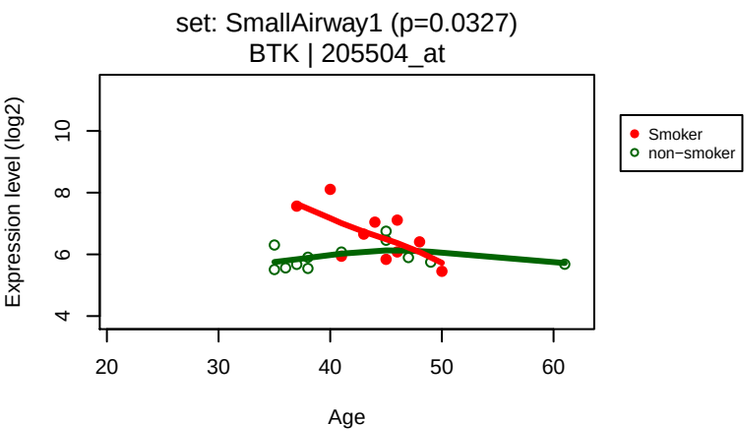
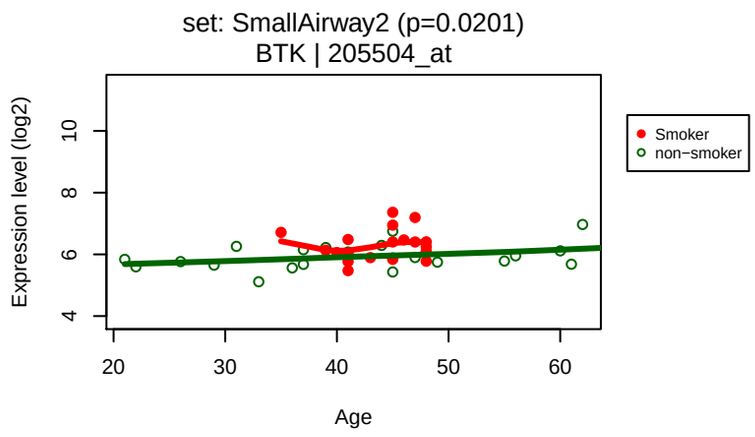
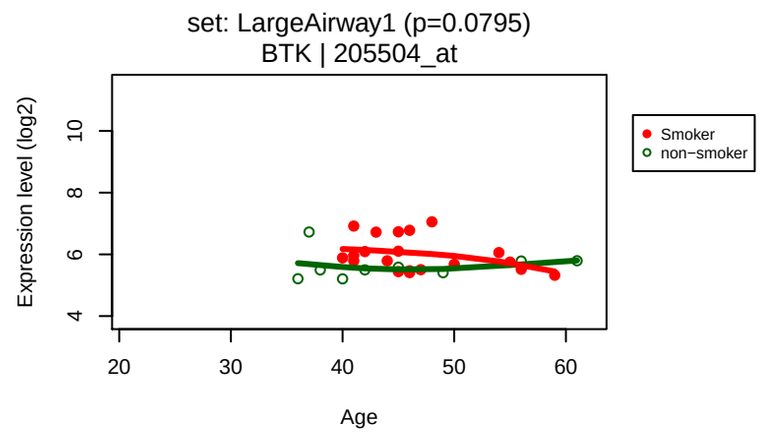

Gene expression of smokers and nonsmokers
gene: BTK | 205504_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0327
p=0.0201
p=0.383
p=0.0795
p=0.6287
p=0.3561
p=0.087
p=0.2215
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

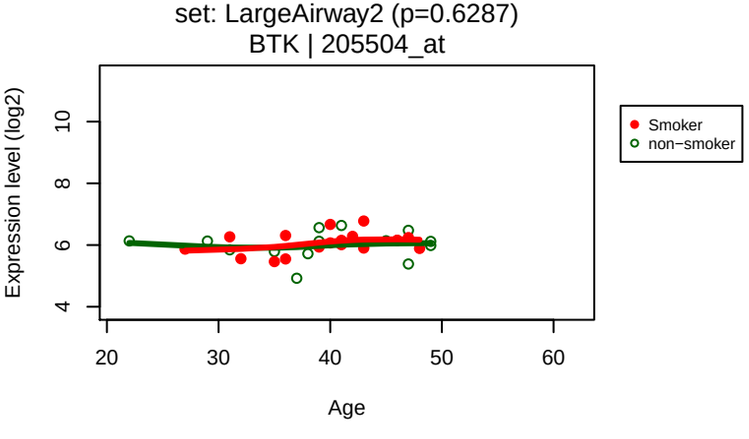

Gene expression ratio (smokers/nonsmokers)
gene: BTK | 205504_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0327
p=0.0201
p=0.383
p=0.0795
p=0.6287
p=0.3561
p=0.087
p=0.2215
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

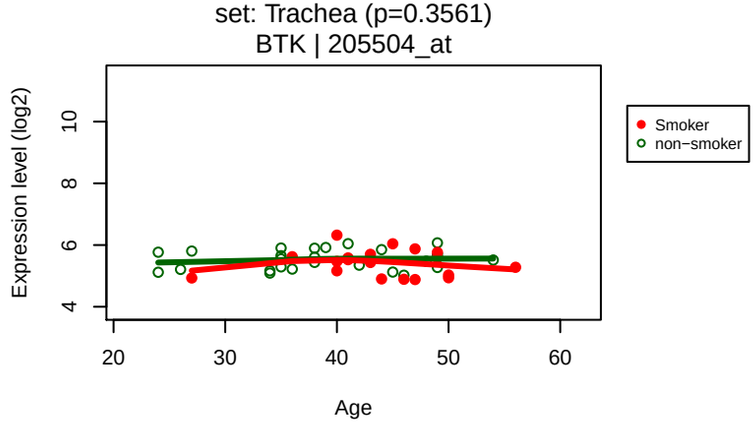

Gene expression ratio in Males
gene: BTK | 205504_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0368
p=0.2514
p=0.3828
p=0.2983
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: BTK | 205504_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3545
p=0.4416
p=0.437
p=0.5126
SmallAirway2
LargeAirway2
Trachea
Macrophage2

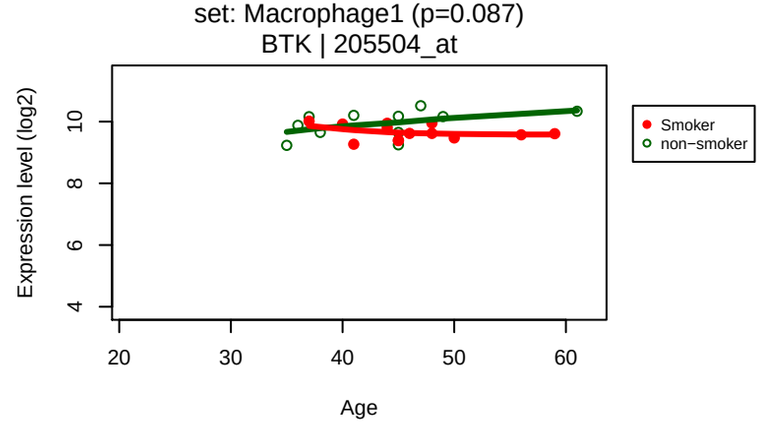

set: SmallAirway1 (p=0.0327)
BTK | 205504_at
Expression level (log2)
Age
Smoker
non−smoker

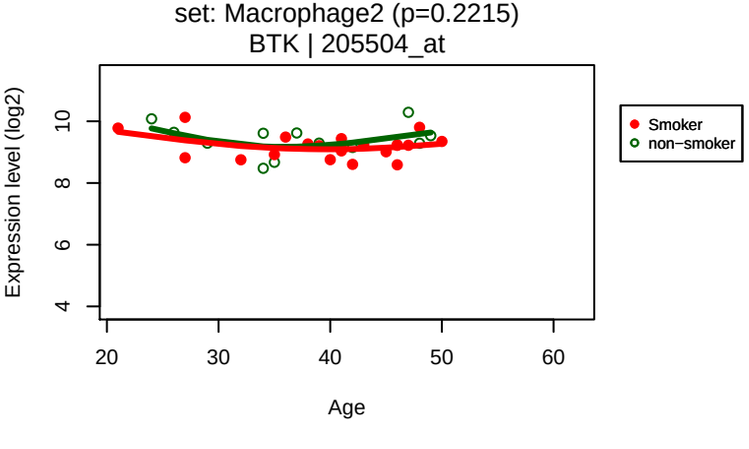

set: SmallAirway2 (p=0.0201)
BTK | 205504_at
Expression level (log2)
Age
Smoker
non−smoker

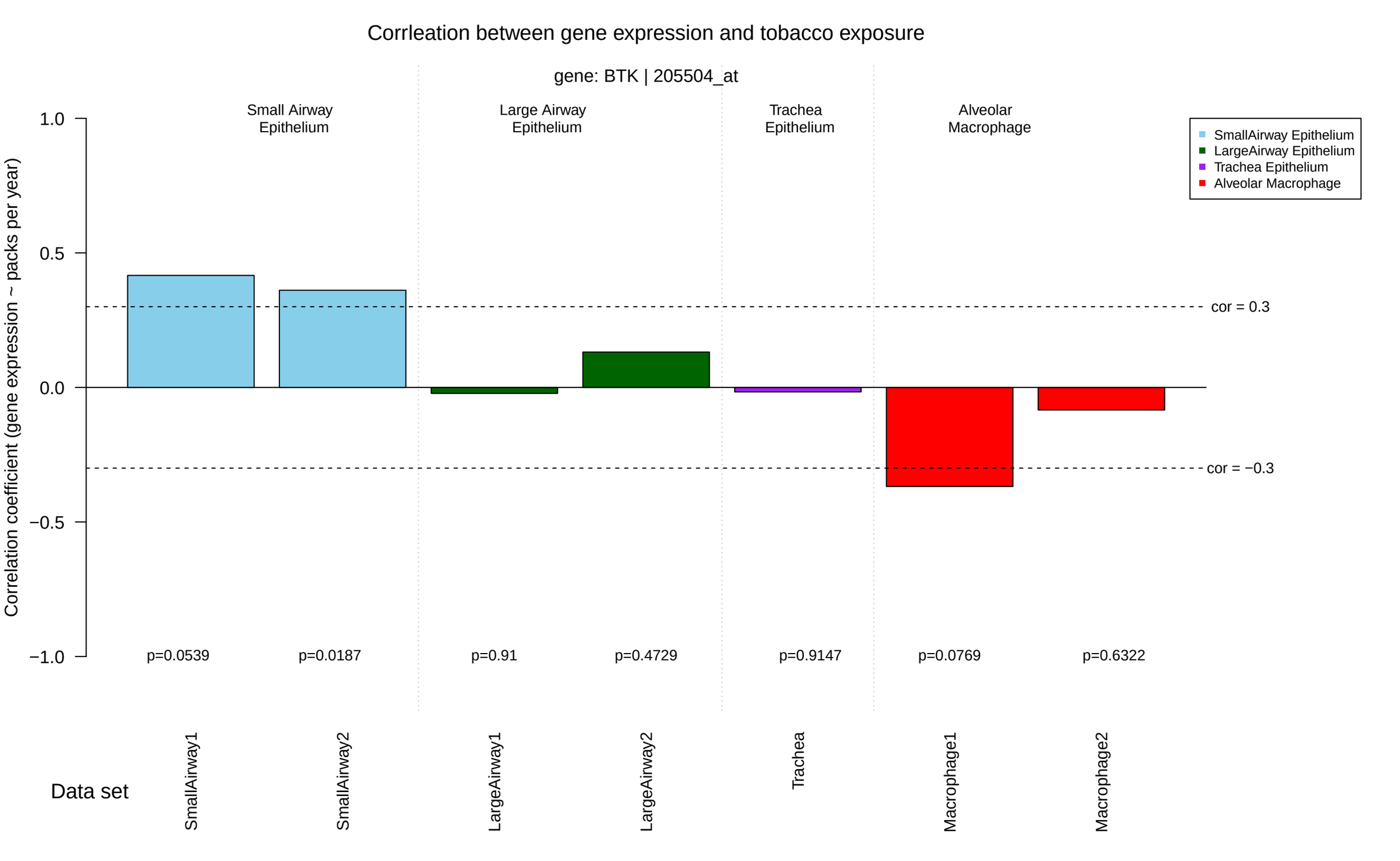

set: LargeAirway1 (p=0.0795)
BTK | 205504_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6287)
BTK | 205504_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3561)
BTK | 205504_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.087)
BTK | 205504_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2215)
BTK | 205504_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: BTK | 205504_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0539
p=0.0187
p=0.91
p=0.4729
p=0.9147
p=0.0769
p=0.6322
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2